Term 4 answer key

Formulas printed for students in every question. Answers with π are rounded to 1 decimal place, as each question says; probabilities are fractions in simplest form; every other answer is exact.

WEEK	SESSION	Q1 A Angles & shapes	Q2 B Measurement	Q3 C Number & percentages	Q4 D Algebra, ratio & rates	Q5 E Number plane, data & chance
1	1	130°	120 m^3	−2	48 and 36	8, 11
	2	85°	48 cm^2	$35	$8m + 8n$	8
	3	56°	12.6 m	24	$3 per L	3/10
2	1	99°	90 cm^2	81	$x = 5$	7
	2	48°	52 cm^2	0.36	5 m/s	1/4
	3	95°	60 cm^2	20%	6	23
3	1	76°	125 cm	$12	1 h 35 min	1/3
	2	72°	54 cm	5.62	$6a + 6$	(3, 5)
	3	150°	55 cm^2	−12	3 : 5	5/8
4	1	61°	44.0 m	63 kg	$x = 13$	24
	2	49°	72 m^3	0.64	$60 and $90	15
	3	116°	30 cm^2	84%	41	13
5	1	110°	30 cm^2	$6	Bag B ($3 per kg)	21/50
	2	60°	196 cm^2	4	$15x − 10$	1/2
	3	150°	34 m	24	$y = 6$	(2, 5)
6	1	70°	120 cm^2	35%	90 km/h	3/4
	2	48°	31.4 cm	10	11	4
	3	102°	2500 mL	60%	8:50 pm	12 °C
7	1	122°	70 cm^2	250	8 : 3	9/25
	2	72°	56 cm^2	$30	$3x + 15$	2
	3	81°	96 cm^3	24	$150 and $90	8, 12
8	1	58°	80 cm^2	12/25	$y = 23$	2/11
	2	42°	28 cm	41	12 km per litre	17/20
	3	123°	31.4 cm	85%	14	12
9	1	115°	91 m^2	−3	2 m/s	15
	2	75°	65 m^2	45 kg	$14 + 7k$	12
	3	140°	18 cm^2	45	$x = 8$	(4, 5)
10	1	110°	350 cm	3/20	35 minutes	1/5
	2	81°	121 m^2	5190	2 : 5	1/2
	3	115°	40 cm	95%	8	22 cm

A MA4-ANG-C-01 complementary and supplementary angles, angles at a point, vertically opposite angles, angles on parallel lines; MA4-GEO-C-01 angle sum of a triangle and a quadrilateral, exterior angle of a triangle

B MA4-LEN-C-01 perimeter, circumference, length units; MA4-ARE-C-01 area of rectangles, squares, triangles, parallelograms and composite shapes, area units; MA4-VOL-C-01 volume of rectangular prisms, volume and capacity units

C MA4-INT-C-01 order of operations with integers; MA4-IND-C-01 powers, square and cube roots; MA4-FRC-C-01 fraction and percentage of a quantity, one quantity as a percentage of another, fractions, decimals and percentages, multiplying and dividing by 10, 100 and 1000*

D MA4-ALG-C-01 substitution, the distributive law; MA4-EQU-C-01 one- and two-step equations; MA4-RAT-C-01 simplifying ratios, sharing in a ratio, rates, best buys, speed; 24-hour time and time differences*

E MA4-LIN-C-01 tables of values, reading points on the number plane; MA4-DAT-C-01 dot plots; MA4-DAT-C-02 mean, median, mode and range; MA4-PRO-C-01 probability, complementary events, experimental probability*

Codes are from the NSW Mathematics K–10 Syllabus (2022). The NESA website could not be reached from the build machine, so **none of these codes was confirmed on the NESA site**; they match the codes used in the Year 9 and Year 10 keys.

* the topic is in this outcome as far as could be checked, but where it sits in the syllabus (or, for 24-hour time, whether it is Stage 3 revision with no Stage 4 code) could not be confirmed.